Viscan Dental OPG AI Assistant

Comprehensive Model Evaluation, Benchmarking & Clinical Metrics Report

Model: Faster R-CNN ResNet-50 FPN V2 (43.27M params, COCO_V1 init)
Benchmark: DENTEX (MICCAI 2023) — Held-Out Test Set (50 images, 182 findings)
Operating Threshold: 0.70 (Val Macro-F1=0.556; Test Macro-F1=0.402)
Live Demo: huggingface.co/spaces/surendra08/viscan-opg
Weights SHA256: e0b49b12f48fc71ec1f583db08f107650659571e7c2b9cbd454dbfed13b46a1f

1. Primary Benchmark Metrics (Held-Out Test Set: 50 images)

mAP @ IoU=0.50	mAP @ IoU=0.50:0.95 (Primary)	mAP @ IoU=0.75 (Strict)	Average Recall (AR@100)
49.6% (0.496)	**32.9%** (0.329)	**38.3%** (0.383)	**68.3%** (0.683)

2. Per-Class Benchmark & Clinical Operating Metrics (Threshold = 0.70)

Pathology Class	COCO AP@50	COCO AP@50:95	Precision	Recall	F1-Score	Test Support	Clinical Reliability
Impacted Tooth	83.3%	54.1%	**76.1%**	**87.5%**	**0.814**	40	**High Reliability:** 87.5% sensitivity with few false alarms.
Deep Caries	45.7%	30.2%	**42.9%**	**18.8%**	**0.261**	32	Conservative detector; 70% of errors are classified as standard caries.
Caries	35.8%	24.4%	**27.7%**	**53.5%**	**0.365**	101	Sensitive screener (captures over half); prone to enamel border FPs.
Periapical Lesion	33.6%	23.0%	**33.3%**	**11.1%**	**0.167**	9	Small sample size (n=9); requires focal loss or oversampling.

3. Test Set Confusion Matrix (Operating Threshold = 0.70)

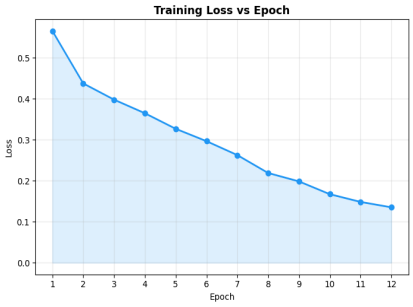
Ground Truth \ Predicted	Background (FN)	Impacted Tooth	Caries	Periapical	Deep Caries
Background (False Positives)	—	10	**124**	2	6
Impacted Tooth	5	**35**	0	0	0
Caries	45	1	**54**	0	1
Periapical Lesion	4	0	3	**1**	1
Deep Caries	12	0	**14**	0	**6**

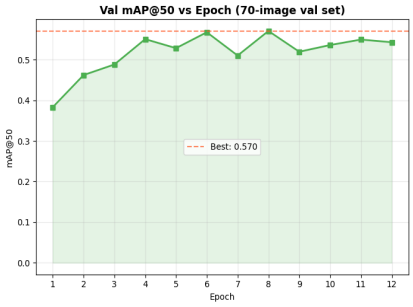
4. Diagnostic & Error Analysis Takeaways

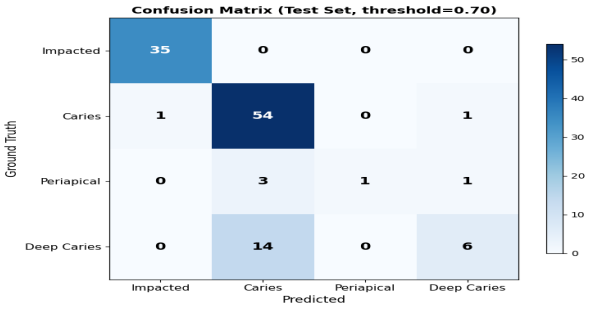
• **Zero Anatomical Cross-Confusion:** Impacted teeth had 0 cross-classifications with carious or periapical lesions (35/40 TP, 5 FN).
• **Boundary Severity Distinction:** Out of 20 total inter-class misclassifications, 14 are deep_caries predicted as caries (severity grading ambiguity near the dentin/pulp border; 6/32 TP as deep_caries, 14/32 localized as caries, 12/32 missed).
• **False Alarm Screener Profile:** 124 of the 142 background False Positives occur on low-contrast caries proposals near interproximal contacts and existing restorations.
• **5 Concrete Test Failure Cases:** Case 1 val_4.png (FP impacted_tooth 0.99), Case 2 val_38.png (FN missed periapical_lesion), Case 3 val_3.png (Misclass deep_caries→caries 0.97), Case 4 val_33.png (FN missed caries), Case 5 val_13.png (FP impacted_tooth 0.99).
• **Resolution Variant Finding:** The 960x1600 variant (bs=1) improved strict mAP@75 (0.410 vs 0.383), while baseline (800x1333,

bs=2) achieved higher primary Val mAP@50:95 (0.360 vs 0.354) and Test mAP@50:95 (0.329 vs 0.313).

5. Empirical Visualizations (Training Dynamics & Dental Test Confusion Matrix)







6. System Specifications, Latency & Regulatory Compliance

Parameter Count: 43,271,528 (43.05M trainable)
Model Architecture: Faster R-CNN ResNet-50 FPN V2 (COCO_V1)
Checkpoint Size: 173.5 MB (173,466,789 bytes)
Training Epochs: 12 (Best: Epoch 8, Val mAP@50=0.570, Val mAP@50:95=0.360)

CPU Inference Latency: 4,618.0 ms mean, 4,915.7 ms P95 (2-vCPU Xeon @ 2.00GHz)
Memory Footprint: 2,926 MB Peak RSS (on 2892x1316 OPG)
Code License: BSD-3-Clause (Torchvision / Commercial Permissive)
Weights License: CC BY-NC-SA 4.0 (DENTEX Non-Commercial Research)

REGULATORY CLINICAL NOTICE: *This AI decision-support system is designed for retrospective and paired clinical evaluation by licensed dental professionals. It does not constitute a primary diagnostic device (FDA SaMD Tier II / MDR Class IIa considerations apply). Outputs are intended as visual assistive prompts for dentist review.*

7. Bonus Track: 3D CBCT Volumetric Segmentation & Nerve Clearance

Explicit Pretrained Model Disclosure: The 3D CBCT bonus pipeline integrates **DentalSegmentator** (Dataset112_DentalSegmentator_v100, Dot et al., *Journal of Dentistry* 2024, Zenodo DOI 10.5281/zenodo.10829675, License: **CC BY 4.0**, nnUNetTrainer__nnUNetPlans__3d_fullres, fold_0, trained on 470 CT/CBCT scans) **pretrained as-is**. Engineering contributions cover NIfTI / 3D DICOM series ingestion, patient tag de-identification, 60 mm³ 3D connected-component post-processing, exact 3D Euclidean nerve-to-root distance computation in physical millimeters (scipy.spatial.cKDTree), Marching Cubes .stl / .glb surface meshing, and browser WebGL integration (<model-viewer>).

Class ID	Anatomical Structure	Voxels (Post-Filtered)	Volume (cm^3)	3D Dice vs. Reference	Mesh Vertices	Mesh Faces
1	**maxilla_upper_skull**	1,476,019	184.50	**0.9768**	181,032	366,752
2	**mandible**	310,556	38.82	**0.9849**	35,515	71,934
3	**upper_teeth**	75,590	9.45	**0.9867**	9,610	19,240
4	**lower_teeth**	62,663	7.83	**0.9745**	8,530	17,068
5	**mandibular_canal**	6,063	0.76	**0.8968**	1,736	3,456
—	**Total / Mean**	**1,930,891**	**241.36**	**0.9639**	**236,423**	**478,450**

• **Clinically Useful 3D Finding (Mandibular Canal Clearance on PostDentalSurgery, 360×360×330 @ 0.5mm):**
  – **Right Hemimandible: min_distance_mm = 0.71 mm** (<1.0 mm high-proximity surgical alert) between closest_lower_tooth_voxel=[119, 151, 89] ([59.5, 75.5, 44.5] mm) and closest_canal_voxel=[118, 151, 88] ([59.0, 75.5, 44.0] mm), bbox_3d_voxel=[108, 141, 78, 129, 161, 99].
  – **Left Hemimandible: min_distance_mm = 1.58 mm** (1.0–2.0 mm close proximity) between closest_lower_tooth_voxel=[235, 132, 75] ([117.5, 66.0, 37.5] mm) and closest_canal_voxel=[236, 132, 72] ([118.0, 66.0, 36.0] mm), bbox_3d_voxel=[225, 122, 62, 246, 142, 85].
• **Measured Memory & Latency (Kaggle NVIDIA Tesla T4 GPU, 14.56 GB VRAM, PyTorch 2.11.0+cu128):**
  – **3D Sliding-Window Segmentation (252 tiles):** 131,407.1 ms (131.41 s) | **Model Init:** 2,131.9 ms
  – **3D Post-Processing + cKDTree Distance:** 5,816.3 ms (5.82 s) | **Marching Cubes Meshing (.stl + .glb):** 938.6 ms (0.94 s)
  – **Peak GPU VRAM Allocated:** 4,936.0 MB (4.94 GB) | **Process CPU RSS Memory:** 2,641.8 MB (+1,270.8 MB delta)
• **Production Architecture for Large CBCT Volumes:** (1) Direct presigned S3/GCS multipart upload with edge DICOM de-identification; (2) Async Redis/Celery task queue returning 202 Accepted + WebSocket progress; (3) Autoscaled NVIDIA L4/T4 GPU workers with pre-warmed VRAM weights and TensorRT FP16 batched sliding-window tiles (cutting 131s to ~20s); (4) Draco-compressed .glb meshes served via CDN to <model-viewer> / vtk.js / three.js.

